

A RESEARCH LOOP: UPDATE THE DESIGN

fixed LLM

Retained code, outcomes and failures guide the next round.

1 Hypothesis

testable change
expected effect

2 Source change

edit task code
feature / policy

4 Selection

score gate
keep / discard

5 Next round

source + scores
failed attempts

validate source

scores + diagnostics

Task contract: writable code | fixed trainer + scorer | roles + budgets

B FEDERATED EXECUTION

PTPC: 1 or 10 simulated clients

Lab 1

D1

loss/grad

Lab 2

D2

loss/grad

Lab K

DK

loss/grad

code + w

loss, grad
count

3a Coordinator

sum objectives
fit weights w

3b Evaluate

dev / CV
fixed scorer

C DOWNSTREAM TASKS

same interface; different adapters

Drug-target interaction

drug + protein $\rightarrow$ binding
central fit / frozen transfer
AUROC, AP

Proteomic efficacy

6h/24h proteomes + drug
federated logistic fitting
AP, AUROC, BCE

Cell policies

ID / phenocopy / CEX
exploratory replay
Top-1, AUC3, AURB3